

# Hands-On Lab: Generative AI for Q and A

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**Estimated time needed:** 30 minutes

## Overview

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In this lab, you will learn how to use **Powerdrill**, a generative AI tool, to gain insights into data through natural language question-and-answer interactions. You will upload a dataset and explore it by asking analytical questions to uncover patterns, relationships, and trends in the data.

## Objectives

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After completing this lab, you will be able to:

1. Sign in on <https://powerdrill.ai/>
2. Upload data set
3. Q and A to know the dataset

## Dataset

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The dataset uses in this lab is generated using a generative AI tool. Keep the Ctrl key pressed and download this dataset from [here](#).

The attributes of the dataset include product ID, title, price, sales rank, brand, category, and availability. These attributes provide product details, such as identification, pricing, popularity (sales rank), brand, category, and availability status.

## Task 1: Signing in on Powerdrill

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Step 1: Click the link below, then select Sign Up if you do not already have login credentials.

<https://chat.powerdrill.ai/>

[illegible]

## Task 2: Upload the dataset

**Step 1:** After logging in, you will see your workspace. Click **Add files** option and select **Choose files**.

The screenshot shows the Powerdrill AI web application interface. The browser address bar displays <https://powerdrill.ai>. The top navigation bar includes links to Microsoft Office Home, Gmail, Zoho People, Dashboard | edX, Timesheet • Clockify, DS\_DA\_DE\_CountSh..., IBM Data Engineer..., (12) Zoho People, Arden meeting, DataAnayst\_Countx..., and Other fa. The left sidebar contains a 'New chat' button, 'Discover', 'Datasets', and a 'History' section. The main content area features a header with the text 'AI for serious data work, unlock 100x efficiency' and 'Loved by over 1,500,000 users globally'. Below this, there are several promotional cards: 'Meet Powerdrill Bloom' (Your AI Data Agents Team!), 'AI Data Report Generator' (Upload your files, click, and your insightful visuals will be ready in minutes.), 'Data Processing Helper', and 'Analysis of Crime Data'. An 'Attach files' modal is open in the foreground, showing two options: 'Upload files' and 'Use existing files'. Under 'Upload files', there is a 'Choose files' button with a red box around it. Under 'Use existing files', there is a 'Select dataset' button. At the bottom of the modal, there is an 'Add files' button with a red box around it, and a 'Start a new chat' button.

**Step 2:** Upload the dataset by selecting the file '**electronics\_dataset.csv**' file from its saved location on your device. Keep the Ctrl key pressed and download it from [here](#) if you still need to do so.

## Task 3: Q and A the data set for better insights

Once the data set is uploaded, you must explore it with natural language-based questions. Start exploring your data by typing the questions (prompts) in the input box.

Analyze with files, quickly gain insights.

  **General mode** Advanced mode  **Send** 

You can begin with the suggested questions as given below.

**Question 1: Identify the products with the highest sales.**

**Prompt:** What are the top-selling products?

What are the top-selling products?

 **Answer** 

 Code  Files  Charts

 **Task Steps** 

Analyzing Conclusions

### Top-Selling Products

#### Total Quantity Sold

 product\_sales\_data.csv

- **Keyboard:** Sold 75 units, making it the top-selling product.
- **Tablet:** Sold 74 units, closely following the keyboard.
- **Monitor:** Sold 65 units, ranking third in sales.
- **Desk:** Sold 62 units, also among the top sellers.
- **External Hard Drive:** Sold 59 units, rounding out the top five.

Now you know that the top-selling products are keyboards, tablets, and so on.

**Question 2: Determine which product has the highest average selling price by calculating the average price for each product.**

**Prompt:** Which product has the highest average price?

Which product has the highest average price?

 Answer 

 Code  Files  Charts

 Task Steps 

Analyzing Conclusions

## Product with the Highest Average Price

### Analysis of Average Price

 average\_price\_data.csv

- **Product with Highest Average Price:** Printer
- **Average Price:** 134.22

**Question 3: Determine the total revenue for the specified time frame.**

**Prompt:** What is the total revenue for the latest two months?

What is the total revenue for the latest two months?

 Answer 

 Code  Files  Charts

 Task Steps 

Analyzing Conclusions

## Total Revenue for the Most Recent Two Months

### Revenue Summary

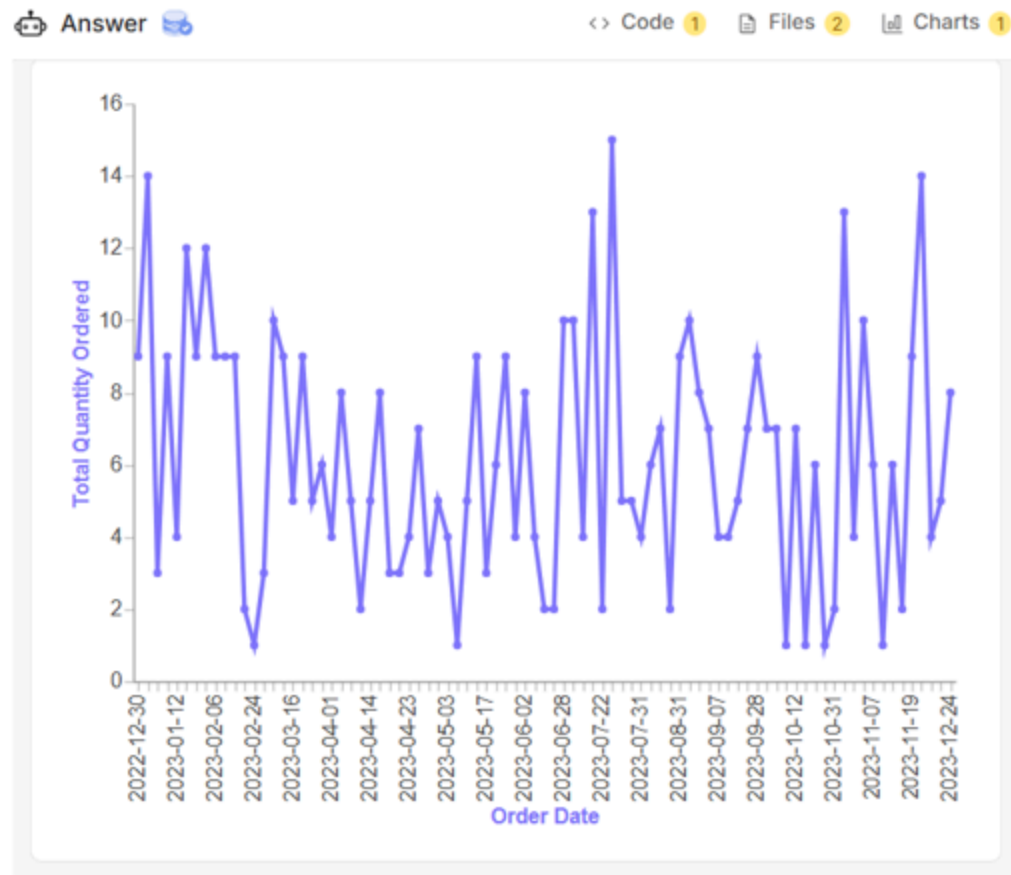
 recent\_months\_revenue.csv

- **October 2023:** \$565.25
- **November 2023:** \$4956.94
- **December 2023:** \$2221.91

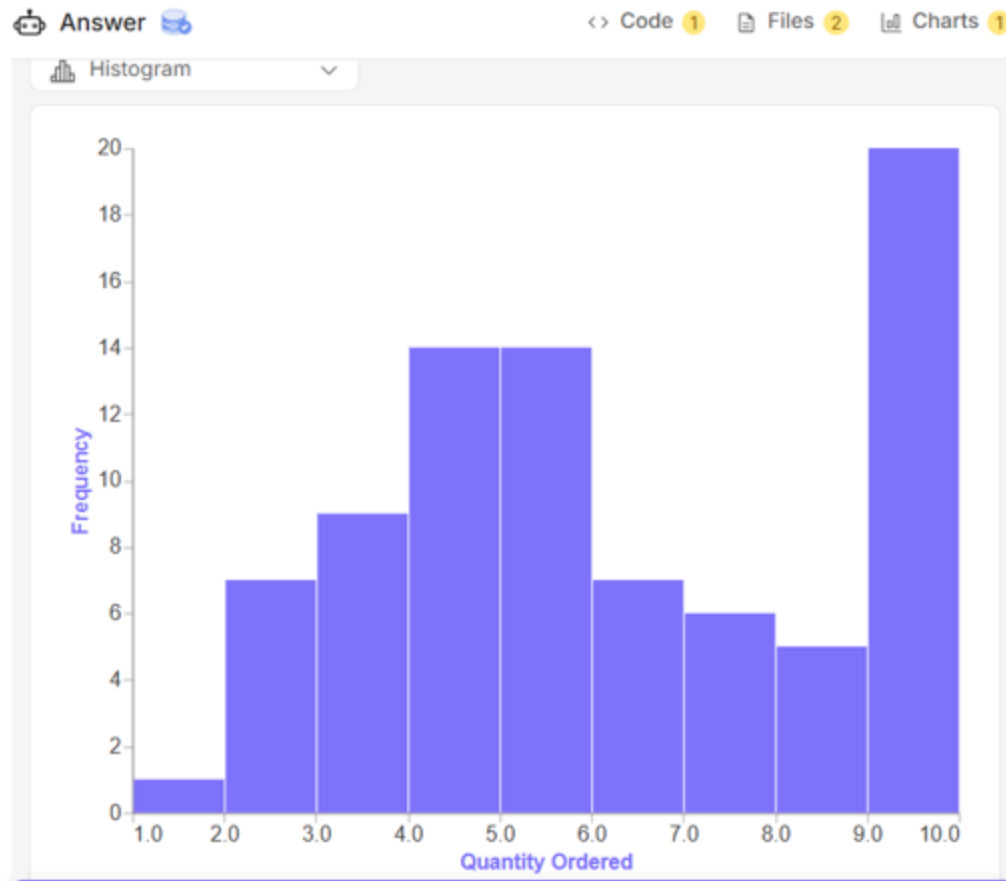
**Question 4: See patterns in the number of items ordered over a period.**

**Prompt:** How has the quantity ordered changed over time?



**Question 5: Determine typical order sizes by histogram of the distribution of order quantities.**

**Prompt:** What is the distribution of order quantities?



**Question 6: Determine the count of orders placed for different quantities of items.**

**Prompt:** How many orders were placed for different quantities of items?

 Task Steps ▾

## Analyzing Conclusions

**Number of Orders for Each Quantity of Items****Orders Count Analysis**

- **Quantity 1:** 10 orders
- **Quantity 2:** 13 orders
- **Quantity 3:** 10 orders
- **Quantity 4:** 13 orders
- **Quantity 5:** 11 orders
- **Quantity 6:** 7 orders
- **Quantity 7:** 6 orders
- **Quantity 8:** 4 orders
- **Quantity 9:** 6 orders
- **Quantity 10:** 10 orders

**Question 7: Explain if there's a relationship between the quantity ordered and the price of each.**

**Prompt:** Are there any correlations between the quantity ordered and price each?

Are there any correlations between the quantity ordered and price each?

 Answer 

 Code  Files  Charts

 Task Steps 

Analyzing Conclusions

## Correlation Analysis

### Correlation Coefficient

 correlation\_coefficient.csv

- **Correlation Coefficient:** The Pearson correlation coefficient between 'Quantity Ordered' and 'Price Each' is **-0.06968**.

**Question 8: Analyze sales patterns to identify the most profitable days.**

**Prompt:** Which day of the week has the highest sales?

Which day of the week has the highest sales?

 Answer 

 Code  Files  Charts

 Task Steps 

Analyzing Conclusions

## Day with Highest Sales

### Sales Analysis

 sales\_by\_day.csv

- **Highest Sales Day:** Tuesday
- **Total Sales Amount on Tuesday:** 11,894.95

**Question 9: Describe the typical order size.**

**Prompt:** What is the average order size?

What is the average order size?

 Answer 

 Task Steps ▾

Analyzing Conclusions

## Average Order Size Calculation

### Calculation Process

- **Total Number of Orders:** There are 100 orders in total.
- **Total Quantity Ordered:** The sum of all quantities ordered is 500.
- **Average Order Size:** The average order size is calculated by dividing the total quantity ordered by the total number of orders, resulting in an average order size of 5.

### Conclusion and Insights

- **Average Order Size:** The average order size is 5, indicating that on average, each order contains 5 items.

## Practice Q and A

1. Look for co-occurrence patterns to find goods commonly bought together.
2. Monitor a product's sales performance to spot trends and seasonality.

## Conclusion

In this lab, you learned how to use generative AI to get insights into data through question answers using the **Powerdrill tool**.

## Author(s)

Pratiksha Verma

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